

Topo-Consynch: Cross-Modal Topological Alignment via the Topological Nyquist Limit and Jiangqiao Topological Extraction Criterion

v1.22 Final Version: From 3D Geometric Perception to 1D Spectral Signal Processing

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Official Release Platform: Hongqiao University Technology Academic Network

July 24, 2026

Philosophical Statement

Idea precedes existence; structure precedes entity.

The TNL-JTEC framework is a discovery of the intrinsic topological constraints of discrete sampling systems, not an invention of empirical engineering heuristics.

Abstract

Persistent homology provides canonical tools to quantify topological structures defined on continuous manifolds and discrete sampled point sets. Nevertheless, the inherent discrete-continuous sampling gap inevitably induces topological aliasing during cross-modal fusion, especially in vision-LiDAR perception systems for autonomous driving and spectroscopic measurement pipelines. We rigorously establish the *Topological Nyquist Limit (TNL)* theorem and the *Jiangqiao Topological Extraction Criterion (JTEC)*. These two core results derive a strict order truncation bound for Taylor expansions of topological quantities: topological features with a persistent lifespan shorter than twice the sampling interval are pure sampling artefacts carrying no intrinsic physical geometric meaning. We further generalize this foundational theory from 3D Riemannian manifolds to one-dimensional smooth manifolds, proposing *Spectral-JTEC* as a topological bandwidth regularizer for spectroscopic signal restoration, denoising and cross-instrument spectral alignment. The entire theoretical system is dimension-agnostic and applicable to all measurement scenarios involving discrete sampling of continuous physical fields. This work establishes a unified geometric sampling paradigm covering robotic perception, analytical chemistry spectroscopy, time-series analysis and acoustic signal processing.

1 Introduction

Autonomous environmental perception and analytical spectroscopy constitute two seemingly disjoint frontiers of modern sensing technology. The former focuses on the

fusion of continuous visual scalar fields $f : \mathcal{M} \rightarrow \mathbb{R}$ and discrete LiDAR point clouds $P \subset \mathcal{M}$; the latter aims to extract molecular fingerprint features from discretely sampled spectral intensity functions $I(\lambda)$. Despite distinct application backgrounds, both fields confront an identical core challenge: *topological aliasing*.

In autonomous driving systems, conventional point cloud registration algorithms merely pursue point-wise geometric matching while neglecting global topological consistency across modalities. Topological Data Analysis (TDA) has evolved into a mature toolkit for extracting multi-scale structural invariants. However, existing TDA pipelines treat topological invariants as post-processed handcrafted features, lacking first-principle theoretical bounds to distinguish physically valid cohomological classes from spurious structures generated solely by discrete sampling.

Likewise in spectroscopy, traditional peak-fitting and peak-picking routines regard every local intensity extremum as a candidate molecular characteristic peak. Under undersampling or strong instrumental noise, such methods yield visually convincing yet chemically spurious narrow spectral peaks. Conventional regularization schemes only impose smoothness or sparsity constraints on the intensity domain, unable to enforce topological validity for recovered signals.

To resolve this fundamental limitation, we construct the Topo-Consynch theoretical framework. We derive the **Topological Nyquist Limit (TNL)**, a fundamental sampling threshold separating robust physical topological modes from unstable aliased components, and formulate the operational **Jiangqiao Topological Extraction Criterion (JTEC)** for cross-modal and cross-dimensional topological fusion. By interpreting the exterior derivative identity $d^2 \equiv 0$ as a universal consistency constraint, we bridge continuous differential topology and discrete computational topology. This manuscript elaborates the theoretical evolution from the 3D perception version series (v1.0–v1.21) to the 1D spectral generalization (v1.22), verifying a universal physical law governing all continuous-to-discrete measurement systems.

2 Theoretical Foundations: The 3D Topo-Consynch Framework

2.1 Dual Cohomological Systems

We model physical scenes as compact C^2 Riemannian manifolds $\mathcal{M} \subset \mathbb{R}^3$ with manifold reach $\tau_{\mathcal{M}}$. Global topological properties are rigorously characterized via de Rham cohomology:

$$H_{dR}^k(\mathcal{M}) = \frac{\ker d^k}{\text{im } d^{k-1}}. \quad (1)$$

To describe multi-scale structural evolution, we adopt persistent de Rham cohomology defined over sublevel-set filtrations $\mathcal{F}_c(t) = \{x \in \mathcal{M} \mid f(x) \leq t\}$.

Given discrete LiDAR point samples $P \subset \mathcal{M}$ with spatial granularity h , we build Vietoris–Rips filtrations $\mathcal{R}(P, \varepsilon)$ and compute discrete persistent cohomology. The bottleneck stability bound $d_B(Dgm_{dR}, Dgm_{\text{discrete}}) \leq 2h$ quantifies the maximum deviation between continuous and discrete topological representations.

2.2 The Topological Nyquist Limit (TNL)

The persistent lifespan ℓ_k of the k -th cohomological class obeys the geometric scaling relation:

$$\ell_k \sim \mathcal{O}(\tau_{\mathcal{M}}^{1/(k+1)}). \quad (2)$$

With increasing cohomological order k , the characteristic spatial scale of topological features shrinks monotonically. We prove the **TNL Theorem**: for 3D spatial sampling tasks, only $k = \{0, 1\}$ classes (connected components and closed loops) possess persistent lifespan $\ell_k \geq 2h$. Higher-order topological features ($k \geq 2$, e.g., enclosed voids) inherently reside below the sampling resolution threshold and cannot be discriminated from random sampling noise.

2.3 Jiangqiao Topological Extraction Criterion (JTEC)

The JTEC criterion acts as a projection operator Π_{JTEC} mapping paired internal/external cohomology spaces onto the admissible low-order topological subspace:

$$\Pi_{\text{JTEC}} : \mathcal{H}_{\text{in}} \times \mathcal{H}_{\text{out}} \rightarrow \mathcal{H}_{\text{low}}. \quad (3)$$

All discrete cohomological components with $k \geq 2$ are truncated systematically as topological aliasing artifacts.

2.4 The Cohomology Consistency Regularizer

Violations of the de Rham fundamental identity $d^2 \equiv 0$ introduced by discrete sampling manifest as unphysical high-order topological modes. We define the curl consistency regularizer:

$$\mathcal{L}_{\text{curl}} = \|d \circ d\omega\|^2. \quad (4)$$

Different from empirical loss penalties, $\mathcal{L}_{\text{curl}}$ serves as a cohomology consistency detector. It suppresses spurious $k \geq 2$ topological signatures while preserving valid low-order topological consensus during gradient-based optimization.

3 Spectral-JTEC: Topological Sampling Theory for One-Dimensional Spectra

3.1 Manifold Interpretation of Spectral Signals

A spectroscopic spectrum is a smooth scalar field $I(\lambda) \in C^\infty(\mathcal{M})$ defined over a one-dimensional compact Riemannian manifold $\mathcal{M}$ parameterized by wavelength axis λ . Discrete spectrometer readings correspond to finite sampling points $P = \{\lambda_i\}_{i=1}^N$, with sampling step $h = \max_i |\lambda_{i+1} - \lambda_i|$. This functorial equivalence between spectral signals and 3D spatial perception enables rigorous dimensional generalization of the entire TNL-JTEC system.

3.2 Persistent Cohomology on 1D Spectra

For intensity function $I(\lambda)$, the sublevel-set filtration $\mathcal{F}_I(t) = \{\lambda \in \mathcal{M} \mid I(\lambda) \leq t\}$ records the birth and death events of topological features:

- H^0 : Connected components, corresponding to spectrally resolved molecular characteristic peaks;
- H^1 : Cyclic topological structures induced by global baseline drift, instrumental ringing effect or closed intensity oscillation patterns.

Genuine cavities and $k \geq 2$ cohomological classes cannot exist intrinsically within 1D topology. Nevertheless, discrete sampling and numerical deconvolution may generate short-lived spurious features mimicking high-order topological modes, which can be efficiently filtered via the JTEC rule.

3.3 Spectral Topological Nyquist Limit Theorem

[Spectral Topological Nyquist Limit] Let h denote the uniform instrumental sampling interval. Any spectral topological feature with persistent cohomological lifespan $\ell < 2h$ is theoretically indistinguishable from sampling-induced topological aliasing and shall not be interpreted as authentic physical molecular signatures. [Remark:] This theorem is a direct dimensional reduction corollary derived from the original 3D TNL theorem (v1.0). No additional axioms are postulated, and it fully inherits the mathematical generality of the foundational framework.

In practical spectral analysis, let W denote the full width at half maximum (FWHM) of a spectral peak. Only peaks satisfying $W \geq 2h$ hold topological validity; all narrower peaks must be discarded regardless of visual prominence.

3.4 Spectral-JTECRegularizer Formulation

The feature retention rule is defined as: a topological birth-death pair (birth, death) is reserved if and only if its persistent lifespan satisfies $\ell = \text{death} - \text{birth} \geq 2h$. The differentiable numerical regularizer is formulated as:

$$\mathcal{L}_{\text{JTEC}} = \sum_i \max(0, 2h - \ell_i)^2. \quad (5)$$

Integrated into the overall spectral reconstruction loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \alpha \mathcal{L}_{\text{sparse}} + \gamma \mathcal{L}_{\text{JTEC}}. \quad (6)$$

Spectral-JTEC supports three core downstream tasks: (1) Topological denoising for sampling-generated pseudo-peaks; (2) Suppression of instrumental baseline artifacts; (3) Cross-device spectral alignment invariant to inconsistent sampling grids.

3.5 Counterintuitive Theoretical Implications

Analogous to the classical Shannon–Nyquist sampling theorem eliminating frequency components exceeding $f_s/2$, Spectral-JTEC rejects all topological features below the $2h$ lifespan threshold. This principle revises conventional chemometric routines that treat every local intensity maximum as chemically meaningful. Spectroscopic applications serve as the first cross-domain empirical verification demonstrating that the TNL–JTEC system is a universal physical sampling law, rather than a tailored geometric engineering heuristic.

4 Iteration Roadmap and Future Work

The Topo-Consynch framework advances following a tiered, scheduled release roadmap without introducing contradictory foundational axioms across versions:

v1.x Series: Establishment of the Universal Topological Sampling Law. **v1.0–v1.21:** Core theoretical construction tailored for 3D autonomous driving geometric perception. **v1.22:** 1D spectral signal generalization (present manuscript). **v1.23:** Extension to time-series vibration and seismic signals (targeting transient event detection via H^0 invariants). **v1.24:** Acoustic and speech signal adaptation (formant extraction based on zero-order cohomology). **v1.25:** Material microscopic image analysis for grain boundary characterization using H^0/H^1 classes. All v1.x iterations only require dimensional adaptation of the core theorem system.

v2.0 Series: Differentiable Numerical Instantiation. Incorporate Discrete Exterior Calculus (DEC), Hodge decomposition theory and $SE(3)$ -equivariant cohomology; develop differentiable persistent homology solvers built upon neural implicit fields to enable real-time on-device inference.

v3.0 Series: Topological Field-Theoretic Unification. Elevate the framework to topological field theory, modeling the bidirectional interaction between the interior moduli space of continuous manifolds $\mathcal{H}_{\text{in}}$ and the exterior moduli space of discrete sampled point sets $\mathcal{H}_{\text{out}}$.

Future Work

Quantitative ablation experiments on public Raman, Fourier-transform infrared (FTIR) and hyperspectral imaging datasets are ongoing, benchmarking Spectral-JTEC against classic chemometric algorithms including Savitzky–Golay filtering and wavelet thresholding. We aim to validate that topological regularization delivers superior chemical interpretability under low signal-to-noise ratio (SNR) measurement conditions.

5 Conclusion

This work introduces Topo-Consynch, a unified cross-modal topological alignment framework rooted in the Topological Nyquist Limit (TNL) and Jiangqiao Topological Extraction Criterion (JTEC). By theoretically proving that only zero- and first-order cohomological classes resist finite-sampling aliasing, we provide a first-principle solution to the ubiquitous topological aliasing problem. The successful generalization to one-dimensional spectroscopic signals verifies the dimension-agnostic property of our theoretical system. This research lays the theoretical foundation for a new generation of geometrically constrained intelligent sensors, bridging high-dimensional differential geometry, analytical chemistry and embodied artificial intelligence.

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